

1. Introduction

Interest rates play a central role in financial markets, influencing the pricing of bonds, derivatives, loans, and a wide range of investment decisions. Since interest rates evolve continuously over time and are affected by numerous economic factors, accurately modeling their behavior has long been an important problem in quantitative finance.

One of the most widely studied approaches for modeling interest rate dynamics is the class of short-rate models, which describe the evolution of the instantaneous interest rate using stochastic processes. Among these models, the Cox-Ingersoll-Ross (CIR) model is particularly significant because it incorporates mean reversion while ensuring that interest rates remain non-negative under appropriate conditions. These properties make it both mathematically appealing and practically relevant for fixed-income valuation and risk management.

The objective of this project is to implement, calibrate, and evaluate the CIR model using historical yield curve data. The model parameters are estimated from observed market data, after which the calibrated model is used to reconstruct the yield curve using only the 3-Month yield as a proxy for the short-term interest rate. The reconstructed yields are then compared against actual market observations to assess the model's predictive capability.

While the CIR model provides a tractable framework for interest rate modeling, it also has several limitations, including its inability to perfectly fit arbitrary yield curve shapes and its restricted flexibility during changing market conditions. To address these shortcomings, this project further explores an extension of the base model and evaluates whether the additional complexity leads to improved out-of-sample performance.

The project follows a complete quantitative modeling workflow consisting of data preprocessing, exploratory data analysis, parameter calibration, yield curve reconstruction, model evaluation, and critical analysis. The ultimate goal is not only to assess the predictive performance of the CIR framework but also to understand its strengths, limitations, and practical applicability in real-world financial modeling.

✓ 2. Dataset Description

The dataset used in this project consists of historical zero-coupon yield observations across multiple maturities. The data is divided into a training set, which is used for model calibration, and a test set, which is used for out-of-sample evaluation.

Each row represents a single observation date, while each column corresponds to a specific maturity on the yield curve. The maturities included in the dataset span from short-term to long-term horizons, allowing the complete term structure of interest rates to be analyzed.

Column Name	Maturity
ZC025YR	3 Months
ZC050YR	6 Months
ZC075YR	9 Months
ZC100YR	1 Year
ZC200YR	2 Years
ZC500YR	5 Years
ZC1000YR	10 Years
ZC2000YR	20 Years
ZC3000YR	30 Years

The training dataset contains 1,976 observations covering all nine maturities. The test dataset contains the 3-Month yield along with selected target maturities that are used to evaluate the predictive performance of the calibrated model.

A key requirement of this project is that, during the prediction phase, only the 3-Month yield is permitted as an input. This yield acts as a proxy for the instantaneous short rate in the CIR framework. Using the calibrated model parameters and the observed 3-Month yield, the remaining points on the yield curve are reconstructed and compared against the actual observed yields.

Before model development, the dataset is examined for consistency, missing values, outliers, and general statistical properties. Understanding the structure and behavior of the data is an essential step in determining whether the assumptions of the CIR model are reasonable for the given market environment.

```
import pandas as pd

train_df = pd.read_csv(
    "https://raw.githubusercontent.com/Grivann/FinClubPS1/main/train_data.csv"
)

test_df = pd.read_csv(
    "https://raw.githubusercontent.com/Grivann/FinClubPS1/main/test_data.csv"
)
```

```
test_3m_df = pd.read_csv(
    "https://raw.githubusercontent.com/Grivann/FinClubPS1/main/test_data_3M.csv"
)

print("Train Shape:", train_df.shape)
print("Test Shape:", test_df.shape)
print("Test 3M Shape:", test_3m_df.shape)

train_df.head()
```

```
Train Shape: (1976, 10)
Test Shape: (495, 6)
Test 3M Shape: (495, 2)
```

	Date	ZC025YR	ZC050YR	ZC075YR	ZC100YR	ZC200YR	ZC500YR	ZC1000YR	ZC2000YR
0	2016-05-19	0.005283	0.005640	0.005846	0.006051	0.006146	0.007912	0.014099	0.028198
1	2016-05-20	0.005286	0.005642	0.005848	0.006053	0.006176	0.007922	0.014179	0.028298
2	2016-05-24	0.005298	0.005651	0.005856	0.006062	0.006228	0.008108	0.014379	0.028598
3	2016-05-25	0.005351	0.005603	0.005809	0.006014	0.006281	0.008323	0.014548	0.028898
4	2016-05-26	0.005354	0.005605	0.005811	0.006016	0.006115	0.007934	0.013937	0.028398

Next steps:

[Generate code with train_df](#)

[New interactive sheet](#)

✓ Data Cleaning

Before proceeding with model development, the dataset was inspected for missing values, duplicate records, inconsistent data types, and potential outliers. Since financial models are highly sensitive to data quality, ensuring the integrity of the yield observations is an important prerequisite for reliable parameter estimation and forecasting.

The following checks were performed:

- Missing value analysis
- Duplicate row detection
- Data type verification
- Summary statistics inspection

- Outlier identification using boxplots

Any issues identified during this stage would be addressed before moving to exploratory analysis and model calibration.

```
import matplotlib.pyplot as plt
import seaborn as sns

print("Training Data Shape:", train_df.shape)
print("\nData Types:")
print(train_df.dtypes)

print("\nMissing Values:")
print(train_df.isnull().sum())

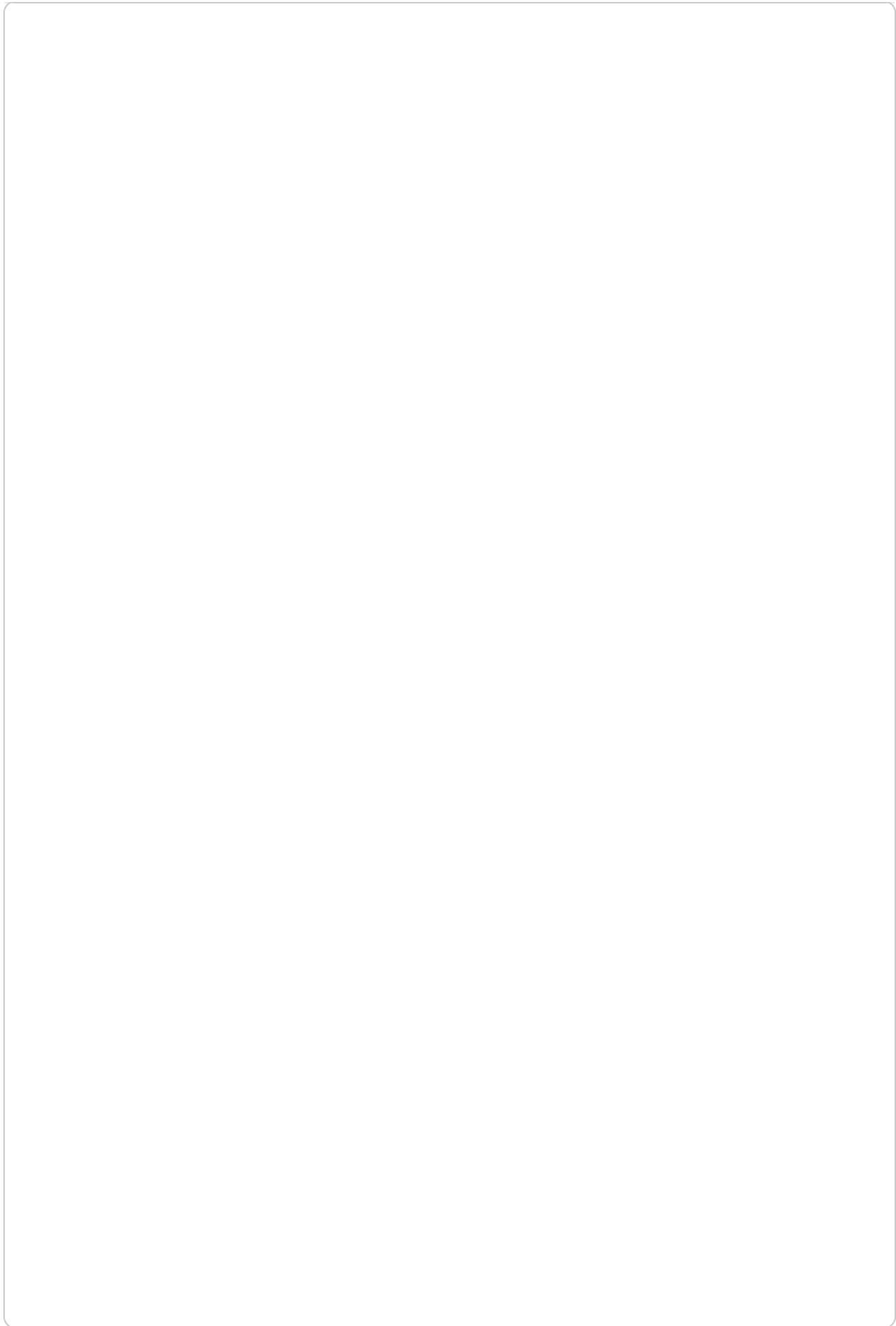
print("\nDuplicate Rows:")
print(train_df.duplicated().sum())

train_df.describe().T

# Check for impossible values
print("Minimum Yield in Dataset:", train_df.iloc[:,1:].min().min())
print("Maximum Yield in Dataset:", train_df.iloc[:,1:].max().max())

plt.figure(figsize=(10,4))
sns.heatmap(train_df.isnull(), cbar=False)
plt.title("Missing Value Heatmap")
plt.show()

plt.figure(figsize=(14,6))
train_df.boxplot(rot=45)
plt.title("Boxplots of Yield Series")
plt.ylabel("Yield")
plt.show()
```



Training Data Shape: (1976, 10)

Data Types:

```
Date          object
ZC025YR       float64
ZC050YR       float64
ZC075YR       float64
ZC100YR       float64
ZC200YR       float64
ZC500YR       float64
ZC1000YR      float64
ZC2000YR      float64
ZC3000YR      float64
dtype: object
```

Missing Values:

```
Date          0
ZC025YR       0
ZC050YR       0
ZC075YR       0
ZC100YR       0
ZC200YR       0
ZC500YR       0
ZC1000YR      0
ZC2000YR      0
ZC3000YR      0
dtype: int64
```

Data Cleaning Summary

The dataset was found to be well-structured and free from missing values or duplicate observations. All columns contained numerical yield data suitable for quantitative analysis. All yields are positive and fall within a realistic range, consistent with the assumptions of the CIR framework.

The boxplots reveal a general increase in median yields as maturity increases, indicating the presence of a predominantly upward-sloping term structure. Short-term maturities exhibit a larger number of outliers, reflecting periods of elevated short-term interest rates caused by changing market conditions. Since these observations are economically meaningful rather than data-entry errors, they were retained in the analysis.

The yield distributions remain positive and exhibit no abnormal patterns, suggesting that the dataset is suitable for calibration of the CIR model without additional preprocessing.

Since the dataset already satisfied the requirements for model calibration, no additional preprocessing was necessary beyond validation and exploratory inspection.

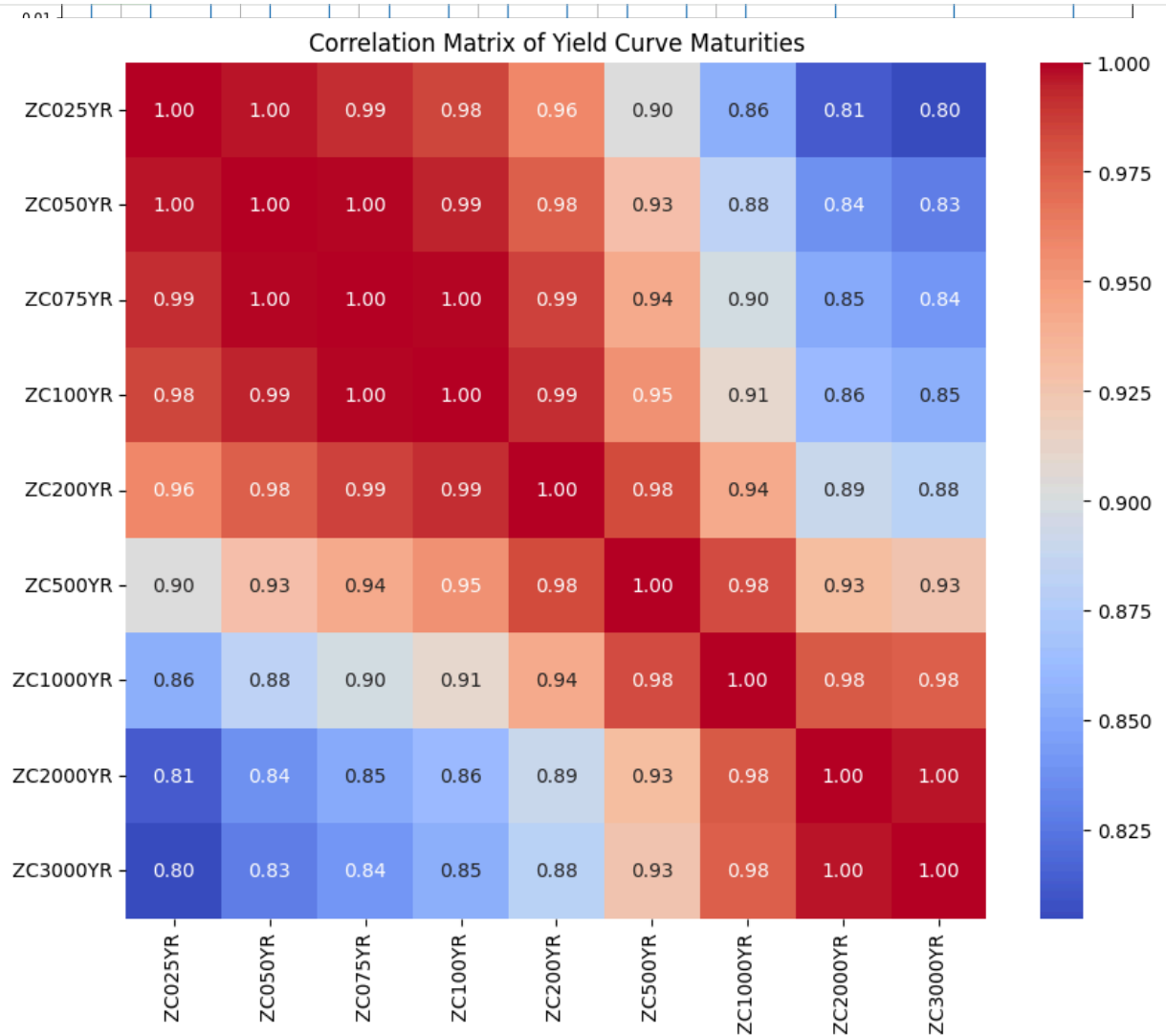
Exploratory Data Analysis

```
import matplotlib.pyplot as plt
import seaborn as sns

corr = train_df.iloc[:,1:].corr()

plt.figure(figsize=(10,8))
sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f")
```

```
plt.title("Correlation Matrix of Yield Curve Maturities")
plt.show()
```



✓ Correlation Analysis

The correlation matrix reveals strong positive relationships across all maturities. Adjacent maturities exhibit particularly high correlations, while correlations gradually decline as the maturity gap increases.

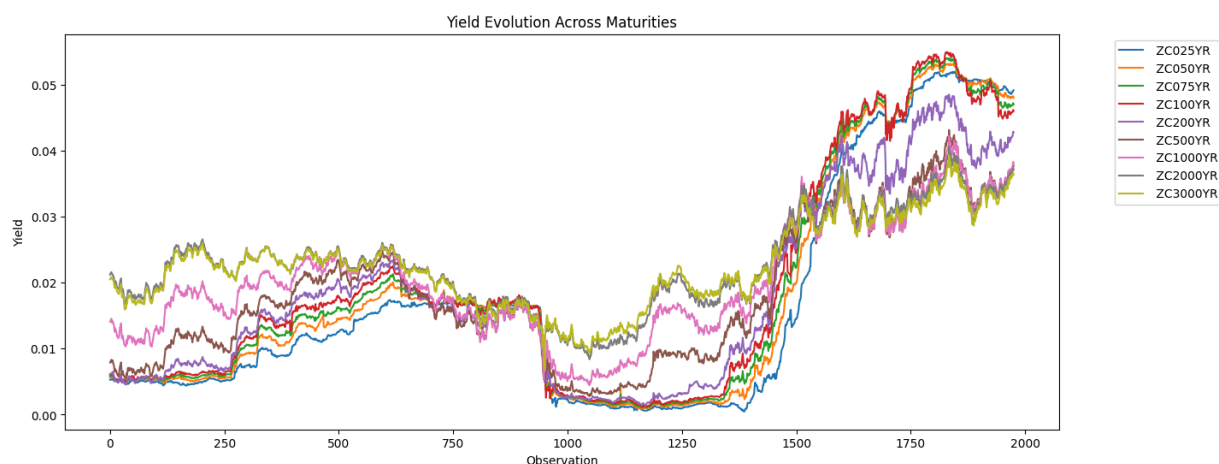
The strong relationship between the 3-Month yield and longer maturities suggests that a significant portion of the yield curve dynamics can be explained by a common

underlying factor. This observation supports the use of the short rate as the primary input in the CIR framework.

```
plt.figure(figsize=(15,6))

for col in train_df.columns[1:]:
    plt.plot(train_df[col], label=col)

plt.title("Yield Evolution Across Maturities")
plt.xlabel("Observation")
plt.ylabel("Yield")
plt.legend(bbox_to_anchor=(1.05,1))
plt.show()
```



✓ Yield Evolution

The yield series move together over time, indicating the presence of common underlying drivers affecting the entire term structure. While the magnitudes differ across maturities, the overall trends remain similar, suggesting strong co-movement throughout the yield curve.

This behavior supports the assumptions of factor-based interest rate models such as CIR.

```
maturities = [0.25,0.5,0.75,1,2,5,10,20,30]

rows = [0,500,1000,1500]

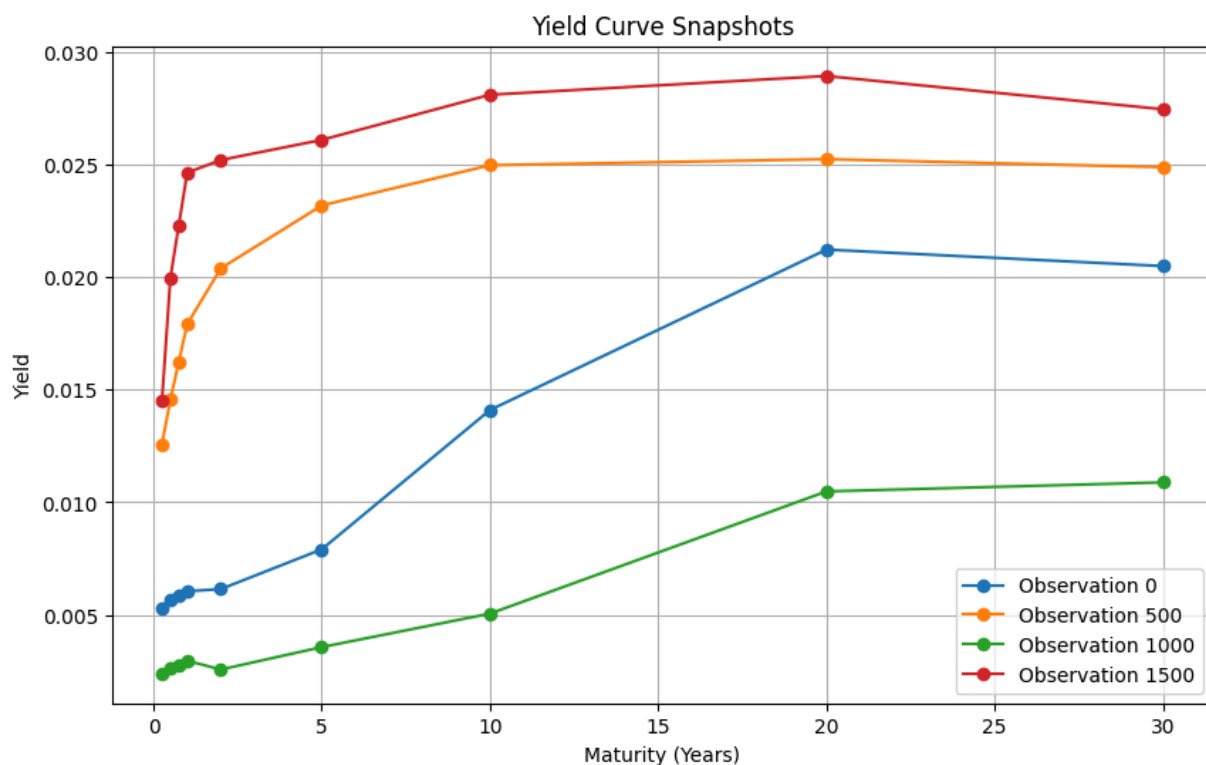
plt.figure(figsize=(10,6))
```

```

for row in rows:
    plt.plot(
        maturities,
        train_df.iloc[row,1:],
        marker='o',
        label=f'Observation {row}'
    )

plt.xlabel("Maturity (Years)")
plt.ylabel("Yield")
plt.title("Yield Curve Snapshots")
plt.legend()
plt.grid(True)
plt.show()

```



Yield Curve Snapshots

Yield curves sampled from different points in time exhibit varying shapes and levels. While many observations display a traditional upward-sloping structure, the slope and curvature of the term structure change over time.

These variations highlight the challenges associated with accurately modeling the entire yield curve using a single-factor framework and motivate the need for model extensions.

EDA Summary

Three key observations emerge from the exploratory analysis:

1. Strong positive correlations exist across all maturities, indicating that movements in the yield curve are largely driven by common underlying factors.
2. Yield series exhibit substantial co-movement through time, suggesting that changes in market conditions affect the entire term structure simultaneously.
3. Yield curve snapshots show predominantly upward-sloping and smooth term structures, although the overall level and shape vary across different periods.

These findings support the use of the 3-Month yield as a proxy for the short rate and provide motivation for applying the CIR framework to reconstruct the remaining maturities.

CIR Theory

✓ Cox-Ingersoll-Ross (CIR) Model

The CIR model assumes that the short-term interest rate follows a mean-reverting stochastic process:

$$dr_t = \kappa(\theta - r_t)dt + \sigma\sqrt{r_t}dW_t$$

where:

- κ : speed of mean reversion
- θ : long-run mean interest rate
- σ : volatility parameter
- W_t : standard Brownian motion

The square-root diffusion term ensures that interest rates remain positive under suitable parameter conditions. The model satisfies the Feller condition:

$$2\kappa\theta \geq \sigma^2$$

which prevents the short rate from reaching negative values.

Start coding or [generate](#) with AI.

Code for CIR Theory

✓ CIR Calibration

Code for CIR Calibration

✓ Yield Curve Reconstruction

Code for Yield Curve Reconstruction

✓ Model Evaluation

Code for Model Evaluation

✓ CIR++ Extension

Code for CIR++ Extension

✓ Model Comparison

Code for Model Comparison

✓ Limitations

Code for Limitations

✓ Conclusion

Code for Conclusion

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